

Spectacle lens design: a review

David A. Atchison

This review considers the present state of the design of spectacle lenses with respect to minimizing the side effects that are associated with their wear. Topics include spherical lens design, aspheric lens design, progressive addition lenses, spectacle lens materials, assumptions of conventional lens design, and practical assessment of spectacle lens design. Future possibilities in the manufacture of spectacle lenses are discussed briefly.

Key words: Image quality, ophthalmic lenses, optical aberrations, spectacle lens design, vision.

Introduction

For the purpose of this paper spectacle lens design refers to the design of lenses to minimize the side effects of wearing corrective spectacle lenses. These side effects include aberrations that reduce image quality; factors that affect spatial perception such as spectacle magnification, distortion, and visual field size; weight; and poor cosmesis. The emphasis of this paper is on the present state; a more detailed description of the development of spectacle lens design is provided in Ref. 1.

Spherical Lens Design

When discussing spherical spectacle lenses, toric lenses are included if the principal sections of their toric surfaces are sections of spheres.

For a spectacle lens to correct an eye for distance vision, the back focal point of the lens must coincide with the eye's far point. If the eye rotates to view distant objects that are not straight ahead of the eye, the far point rotates about the eye's center of rotation to map out the far point sphere (Fig. 1.). Ideally light would be focused on the far point sphere. This is not possible, and thus designers try to reduce important aberrations.

When a light pencil is incident obliquely on a lens, it is refracted differently in different meridians. Light that is refracted in the tangential section, which contains the optical axis and the pupil ray, comes to a different focus than light that is refracted in the

sagittal section at right angles to it. The difference between the tangential focus and the far point sphere is measured by the tangential power error, and the corresponding difference between the sagittal focus and the far point sphere is measured by the sagittal power error. The difference between the tangential and sagittal power errors is called oblique astigmatism, and the average of the two errors is called the mean oblique error.

Different weightings to the amounts of errors can be used by lens designers, such as eliminating oblique astigmatism and leaving a small residual mean oblique error or eliminating mean oblique error and leaving a small residual oblique astigmatism (Fig. 2). As well as the tangential and sagittal power errors the only other aberration that is generally considered to affect foveal image quality is the transverse chromatic aberration. This particularly affects the light that is refracted in the tangential meridian (Fig. 3). It can only be affected by the choice of lens material. When it is taken into account in lens design, lenses are well corrected for the tangential power error.¹

There is only one major design variable that influences the tangential and sagittal power errors of spherical spectacle lenses: the shape or the bending factor. This is determined by the choice of the base curve, which is the surface power from which the other surface power is calculated for one or more lenses. The base curve, which is selected for a given lens or range of lens powers, is that considered to be the best compromise over the likely range of variables, which include vertex distance, pantoscopic tilt, object distance, and off-axis angles.

Lenses that are designed to optimize foveal image quality, as discussed above, when the eye rotates behind them are referred to as corrected curve lenses in the United States and best form in the United Kingdom.

The author is with the Centre for Eye Research, School of Optometry, Queensland University of Technology, GPO Box 2434, Brisbane Q 4001, Australia.

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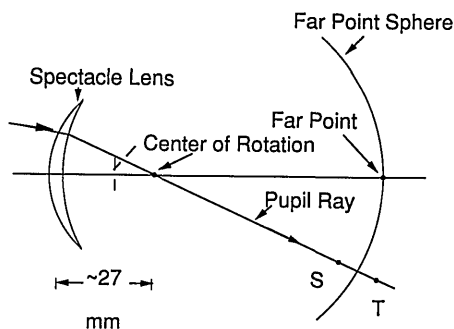


Fig. 1. Basis for conventional spectacle lens design.

By using approximate ray-tracing methods in which lenses are assumed to be thin and only third-order aberrations are considered, the tangential and sagittal power errors are found to be quadratic functions of the back (or front) surface power. Accordingly, for each error type and a particular combination of other parameters, there are either two solutions of the base curve, no solutions, or two equal solutions. The solutions lie on a curve that is an oblique ellipse (Tscherning ellipse) when power is plotted against the back surface power. For oblique astigmatism the lens power limits to solutions are approximately -23 to $+7$ diopters (D) (Fig. 4). Most spectacle lenses are designed to lie near the top Ostwalt section of the ellipse rather than near the bottom Wollaston section, where the lenses are extremely curved and are less cosmetically appealing.

For much of this century manufacturers have argued about the relative merits of different corrected curve designs. They have sometimes disagreed about lens designs that vary in base curve by <1 D and

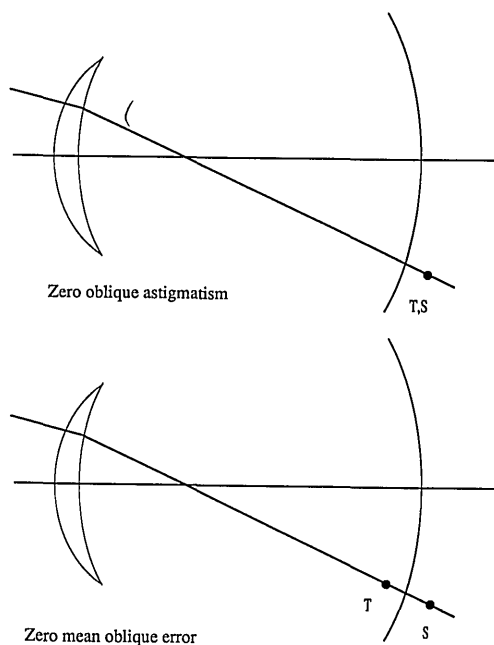


Fig. 2. Zero oblique astigmatism with residual mean oblique error (top) and zero mean oblique error with residual oblique astigmatism.

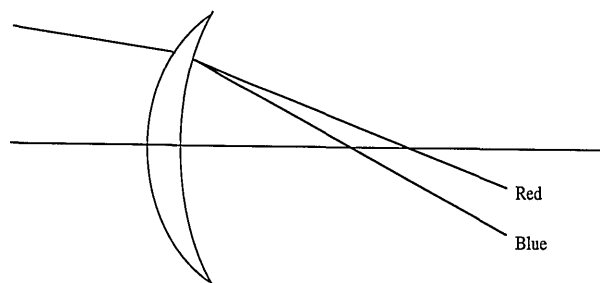


Fig. 3. Transverse chromatic aberration with spectacle lenses.

whose peripheral aberrations differ by $\sim \frac{1}{8}$ D. This amount is insignificant when one considers that this is less than the reliability of subjective refraction. Patients do not often complain of poor vision when looking through the periphery of their lenses! The effects of the base curve on image quality are important, but they can be varied considerably without detrimental effects, particularly in low-power lenses.

Aspheric Spectacle Lens Design

When designing spherical lenses there is only one major design variable: the shape (bending) factor. The lack of design variables means that the designer has little hope of reducing aberrations simultaneously. Aspherizing one or both of the surfaces of lenses allows the correction of off-axis powers in high-positive power lenses where this would be otherwise impossible and enables other factors to be considered simultaneously with off-axis power error correction over the total range of lens powers.

This can be investigated by aspherizing one surface of the lens as a conicoid, which is the surface of revolution of a conic section about an axis of symmetry of the conic. A conicoid that passes through the

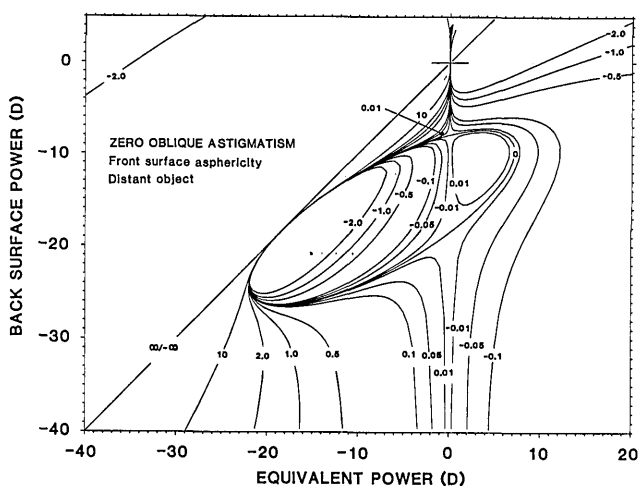


Fig. 4. Back surface powers that provide zero oblique astigmatism as a function of the equivalent power for various conic asphericities applied to the front surface. The curve marked with a zero (0) is the Tscherning ellipse for spherical lenses. If Q is negative, the front surface flattens away from its vertex. If Q is positive, the front surface steepens away from its vertex. The lens refractive index, 1.5. The center-of-rotation distance is 27 mm, distance vision.

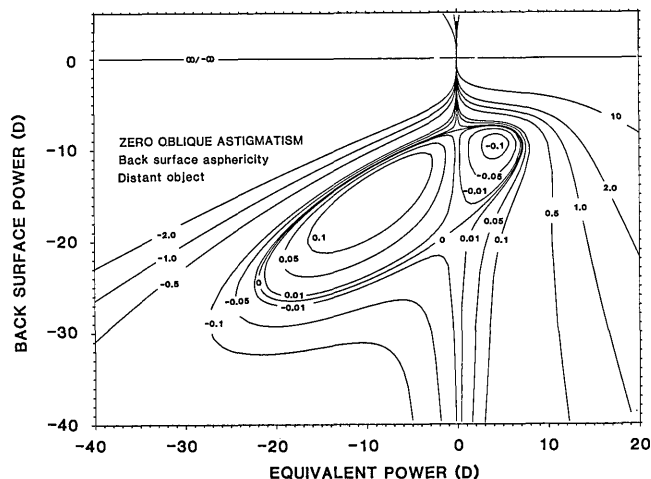


Fig. 5. Back surface powers that provide zero oblique astigmatism as a function of equivalent power for various conic asphericities applied to the back surface. Other details are as for Fig. 4.

point (0, 0, 0) and is rotationally symmetric about the z axis (optical axis) can be expressed in the form

$$z = \frac{cp^2}{1 + [1 - c^2(1 + Q)p^2]^{0.5}},$$

where z is the sag of the surface, c is the vertex curvature, and $p^2 = x^2 + y^2$. Q is the asphericity of the conicoid. If Q is negative, the surface flattens away from the vertex. If Q is positive, the surface steepens away from its vertex; if Q is zero, the surface is spherical.

By using the approximate ray-tracing methods that I referred to above, when one surface of a spherical lens is made into a conicoid, a conventional error such as oblique astigmatism becomes a cubic function of the back (or front) surface power^{2,3} (Figs. 4 and 5). Since a cubic function has at least one root, at any lens power there is now always at least one value of the back surface power that eliminates the error.

Until intraocular lenses became commonly used in developed countries, the most important application for ophthalmic aspheric surfaces was for high-positive-power lenses with approximate powers of +10 to +16 D that are used for the correction of aphakia. Oblique astigmatism and mean oblique errors cannot be substantially reduced in these powers with spherical surfaces. For such lenses the required front surface asphericity is negative (Fig. 4), which indicates that the front surface flattens toward the edge, which has the additional advantages of enabling the lens to be thinner and lighter, reducing spectacle magnification, reducing distortion, and increasing the visual field size.

Aspheric lenses for the correction of aphakia were first introduced by Zeiss in 1909. Throughout the 1970's and 1980's there have been many lens designs that take into account such factors as spectacle magnification, thickness, and visual field size in addition to the reduction of off-axis aberrations.^{4,5} There has been little design development of aphakic lenses

in the last few years because of the widespread use of intraocular lenses.

In recent years, however, asphericity has been introduced into the moderate power range where bending alone can be used to reduce substantially the off-axis power errors. An additional degree of freedom has been used to reduce thickness and hence improve cosmesis without too much sacrifice of the off-axis image quality. Companies such as Essilor, Hoya, Rodenstock, and Sola have introduced aspheric lens series. Most companies use aspheric surfaces that are more complicated than conicoids, for example, of the form

$$z = z_c + A_4p^4 + A_6p^6 + A_8p^8 + A_{10}p^{10},$$

where z_c is now a conicoid and the other terms are referred to as figuring terms. $A_4, A_6, \dots$, are coefficients. The image quality is given an important priority out to a particular diameter, beyond which variations in the surface shape are for the purpose of reducing lens thickness. Manufacturers are using high-index materials to improve further the cosmetic appearance that is obtained with aspheric surfaces.

For moderate-positive-power lenses a reasonable approach is to reduce the surface powers from the Ostwalt forms and apply negative asphericity to the front surface (Fig. 4). This can maintain off-axis image quality while reducing the center thickness, weight, and spectacle magnification, but distortion may be increased. An example for +6-D power is given in Fig. 6. A best form spherical lens has a center thickness of 7.0 mm and an edge thickness of 1.0 mm at a 60-mm diameter. The optimum base curve is +10 D, which gives low sagittal and tangential power errors. I designed an aspheric lens with a +7-D base curve, a front surface conicoid asphericity of $Q = -1.5$, and a front surface A_{10} figuring coefficient of $-10^{-15} \text{ mm}^{-9}$. Aspherizing makes it possible to reduce the center thickness from 7 to 5.5 mm and reduce weight by 25%. The improvement in cosmetic appearance is obvious. Spectacle magnification (SM) is re-

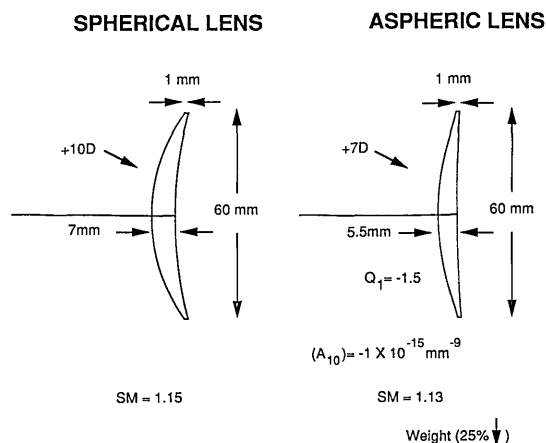


Fig. 6. Cross section of two +6-D spectacle lenses. One is a spherical lens and the other is with a figured conicoid front surface. The refractive index is 1.5.

duced slightly from 15% to 13%. Distortion is slightly higher, but there is also a slightly larger visual field (Fig. 7). The off-axis performance is similar for the two lenses out to a 27.5° angle of rotation, beyond which that for the aspheric lens deteriorates rapidly (Fig. 8).

For positive-power lenses it is also possible to reduce the surface powers and provide positive asphericity on the back surface (Fig. 5).

For moderate-negative-power lenses the most useful approach is to reduce the surface powers from the Ostwalt forms and apply negative asphericity to the back surface (Fig. 5). Without reducing off-axis imagery, edge thickness and weight can be decreased, but again there will be an increase in distortion. An example for a -6-D power is given in Fig. 9. A best form spherical lens has a center thickness of 2.0 mm and an edge thickness of 8.0 mm at a 60-mm diameter. The optimum base curve is +2.5 D, which gives low sagittal and tangential power errors. I designed an aspheric lens with a +1-D base curve, a back surface conicoid asphericity of $Q = -1.3$, and a back surface A_{10} figuring coefficient of $-2 \times 10^{-15} \text{ mm}^{-9}$. The lenses are shown here. Aspherizing reduces the edge thickness from 8.0 to 6.1 mm and reduces weight by 8%. The improvement in cosmetic appearance is obvious. Distortion is slightly higher (Fig. 10). The off-axis performance is similar for the two lenses out to a 27.5° angle of rotation, beyond which that for the aspheric lens deteriorates rapidly (Fig. 11).

For negative-power lenses it is also possible to reduce the surface powers and provide positive asphericity on the front surface (Fig. 4).

The optical performance of aspheric lenses is more susceptible than that of spherical lenses to incorrect fitting. Most aspheric lens manufacturers emphasize that the pantoscopic tilt be matched to the fitting height of the lens center so that the center of rotation of the eye is on the optical axis of the lens.

Progressive Addition Lenses

The main activity in multifocal spectacle lens design has been in the field of progressive addition lenses. Most of the work in multifocal lenses with discrete powers has been concerned with technical problems; the designs of most bifocals and trifocals are intended to reduce the distance off-axis power errors.

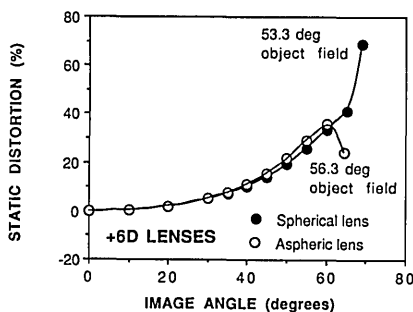


Fig. 7. Static (stationary) distortion of +6-D spectacle lenses, as shown in Fig. 6, as a function of the image angle. The entrance pupil of the eye is 15 mm behind the lens back vertex.

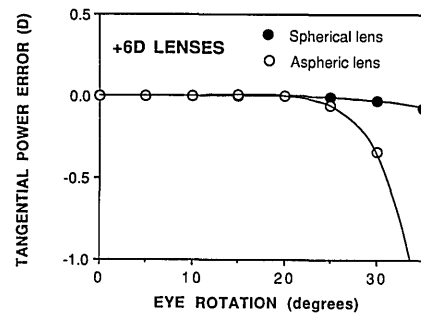


Fig. 8. Tangential power errors (D) of +6-D spectacle lenses, as shown in Fig. 6, as a function of eye rotation. For clarity, the sagittal power errors have been omitted because they do not change as quickly as the tangential power errors. The center-of-rotation distance behind the lens back vertex is 27 mm.

The purposes of progressive addition lenses are to give the presbyope clear vision at a range of viewing distances without image jump and a detrimental cosmetic appearance caused by obvious boundaries between the focal zones of the lenses. They have a corridor of progressively increasing power that is provided by a nonrotationally symmetrical aspheric surface. The design problem is to reduce or to move to less important areas of the lens introduced astigmatism and distortion so that they interfere minimally with vision. The distortions cannot be removed completely and may produce uncomfortable swimming of the visual field when a patient looks through the periphery below the distance portion.

There are dozens of designs available that are based on a few different principles. The main change in recent years has been a move from hard designs in which the aberrations are confined to relatively small regions of the lens but change quickly and can be quite high to softer designs in which the aberrations are smaller but are spread over a larger area of the surface, thus restricting the size of the visually important areas. Results of wearer trials have supported this change.⁶⁻⁹ There is also concern about the vertical symmetry of the design, as the rotation of lenses to make them right or left lenses can create

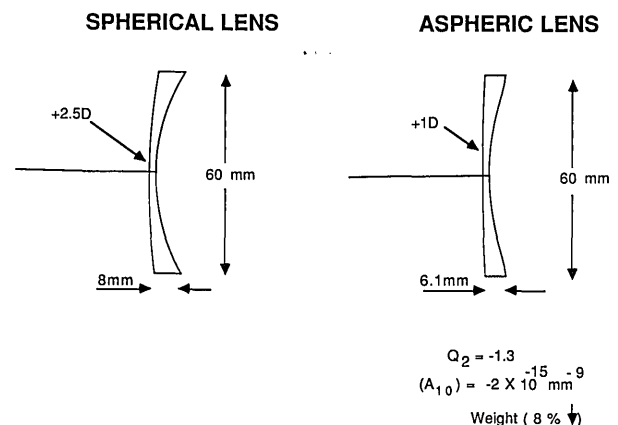


Fig. 9. Cross section of two -6-D spectacle lenses. One is a spherical lens, and the other is with a figured conicoid back surface. The refractive index is 1.5.

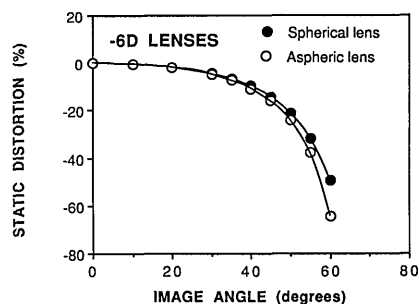


Fig. 10. Static (stationary) distortion of -6 -D spectacle lenses, as shown in Fig. 9, as a function of the image angle. The entrance pupil of the eye is 15 mm behind the lens back vertex.

different aberrations and induce differential prismatic effects when looking through the corresponding points of the two lenses.^{10,11} Hence some companies make lenses as separate right and left lenses. There is also a trend toward having a range of designs available to suit the majority of wearers. Some manufacturers design differently for different addition powers.

Many factors contribute to lens performance and wearer acceptance, not all of which are presently well understood. These include occupation, personality, gender, refractive error, power addition, and previous wearing history.¹² Other papers in this *Applied Optics* issue cover the topic of progressive-power lenses in greater detail.

Spectacle Lens Materials

For many years the standard material for spectacle lenses was ophthalmic crown glass (refractive index $n = 1.523$ at 587.6 nm, specific gravity $sg = 2.54$, constringence $V \sim 60$). The use of hard resin CR39 ($n = 1.499$, $sg = 1.32$, $V \sim 60$) has overtaken that of glass on the market in most developed countries because of the former's flexibility with respect to design and tinting, its low density, and its high impact resistance. Protection from UV radiation is good. Higher-index plastic materials ($n = 1.56$ – 1.60) have been developed recently for use in lenses of moderate to high powers, often with aspheric surfaces. Relative

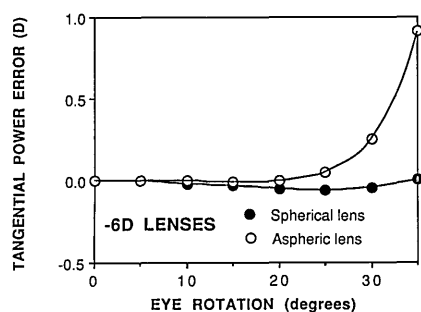


Fig. 11. Tangential power errors (D) of -6 -D spectacle lenses, as shown in Fig. 9, as a function of eye rotation. For clarity the sagittal power errors have been omitted because they do not change so quickly as the tangential power errors. The center-of-rotation distance behind the lens back vertex is 27 mm.

to CR39 they give a reduced center thickness for positive-power lenses and a reduced edge thickness for negative lenses. However, they suffer from low constringence ($V \sim 35$ – 40), which worsens transverse chromatic aberration.

Another plastic material that is being used increasingly because of its extreme high impact resistance is polycarbonate. However, it should be realized that this result applies only to blunt objects; it is vulnerable to penetration by sharp objects. It has a lower specific gravity ($sg = 1.20$) and a higher refractive index than CR39 ($n = 1.58$ – 1.59) but a lower constringence ($V \sim 30$). Improvements have continued in recent years in higher-index glass materials with reductions in specific gravity and increases in constringence. The service to patients who require higher-power lenses and for whom cosmetic appearance is of the utmost concern is thus being improved.

Assumptions of Conventional Spectacle Lens Design

Apart from providing the position of the effective stop, the eye does not form part of the conventional ray-tracing schema. There is an assumption that is inherent in the method of analysis, which is that pupil size can be ignored because it is small. Aberrations that are highly dependent on aperture stop diameter, spherical aberration and coma, are ignored. Foveal monochromatic image quality is assumed to be that which the eye would have if suffering from refractive errors of the same longitudinal magnitude as the lens off-axis power errors. Vignetting by the lens surfaces is also ignored.

To assess the assumption in single-vision lenses, I traced rays through lenses and eyes with finite pupil sizes, which rotated about their center of rotation so that the fovea was aligned with off-axis points of interest.¹³ The eyes had typical chromatic aberration and spherical aberration. Following ray tracing, wave aberrations and modulated transfer functions (MTF's) were used as image quality criteria.

It is indeed appropriate to ignore the aberrations of the lens, which are highly dependent on pupil size. This is illustrated in Table I which shows wave aberrations at the edge of 8-mm pupils for two $+6$ -D lenses, which correct oblique astigmatism. Spherical aberration is negligible at 0.37 wavelengths and 0.03 wavelengths compared with the field curvature (corresponding to the mean oblique error) of -6.1 wavelengths. Lens coma is high at large pupils with some designs of lenses, such as -4.7 wavelengths with the highly aspheric $+6$ -D lens, but this has little effect on

Table I. Wave Aberration Components in Wavelengths of Two $+6$ -D Lenses that Correct Oblique Astigmatism at 30° Rotation (8-mm Pupil)

	$+12$ -D Base Curve, Spherical	$+6$ -D Base Curve, Aspheric $Q_1 - 4.75$
W_{20} field curvature	-6.06	-6.08
W_{22} oblique astigmatism	-0.05	-0.02
W_{31} coma	$+0.18$	-4.67
W_{40} spherical aberration	$+0.37$	$+0.03$

foveal image quality even for large pupils. This is shown in Fig. 12 in which the MTF for an eye that is corrected with the +6-D aspheric lens is plotted against spatial frequency. The image quality is much worse for 30° off-axis rotation of the eye than it is on-axis, but if the off-axis field curvature is corrected, there is a negligible difference between the two. Thus the tangential and sagittal errors can be equated with refractive errors of the same magnitude. Vignetting is only important near the edge of a lens, so that distortion and visual field size determination using conventional ray tracing is adequate.

In general the assumption of conventional spectacle lens design seems to be valid.¹³ It is possible that the use of complex, rapidly changing lens surfaces may produce considerable coma or comalike aberrations, and the designers of progressive addition lenses are starting to consider this.¹⁴

Practical Assessment of Spectacle Lens Design

Laboratory Assessment

To measure off-axis sagittal and tangential aberrations and distortions of single-vision lenses, simple optical bench apparatus has been developed and conventional focimeters have been modified so that lenses rotate about a point corresponding to the center of rotation of the eye. To assess surfaces, topographical instruments and modified radiuscopes have been employed.

The main interest in practical assessment concerns progressive addition lenses. Many positions on these must be considered because of the complexity of the progressive additional surface. Possible assessments include the following:

- (1) Use of a conventional focimeter and an attachment to rotate a lens about a point that is typically 27 mm behind the lens back vertex to measure spherical and cylindrical powers and prism, e.g., Refs. 15 and 16.

- (2) Automatization of the measurements in method (1) by using an automated focimeter. The

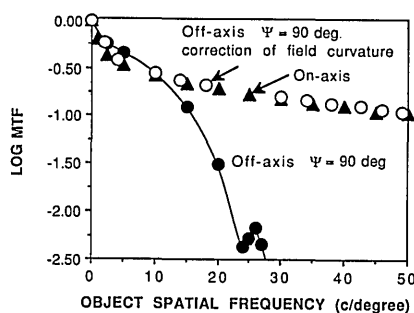


Fig. 12. Monochromatic MTF's as a function of the object spatial frequency for an aspheric +6-D lens that is corrected for oblique astigmatism, in combination with an eye model. Results are shown for on-axis viewing, 30° off-axis viewing with target orientation at 90° to the tangential plane (this target orientation being that most affected by coma), and 30° off-axis viewing with the target orientation at 90° to the tangential plane and a focal shift to correct the lens field curvature. See Ref. 13 for further details.

focimeter and attachment are under computer control. The lens is either rotated about a center-of-rotation position of the eye¹⁷ or about the center of curvature of the back surface.¹⁸

- (3) Surface topographical measurements of the progressive surface, mathematical curve fitting to the surface, and subsequent theoretical ray tracing.¹⁴

- (4) Use of devices for measuring surface curvatures across the progressive surface.¹⁹

- (5) Interferometric techniques that use a standard lens surface for comparison with test surfaces.²⁰

The first method provides a satisfactory simulation of the eye rotating behind the lens with a distant target and is easy and inexpensive to develop. However, this manual method is slow and is limited in precision and reproducibility. The second method may not necessarily give the same results as method (1) because of the different principles of conventional and automated focimeters, but for the best form lenses of near plano distance power the results are similar.^{21,22} The third method requires sophisticated machinery and software to describe the progressive surface appropriately, but is flexible in that a number of parameters such as working distance, pantoscopic tilt, and lens power may be readily manipulated. Also various optical characteristics may be demonstrated readily. The final method appears to be of particular value in quality control.

Recently an ad hoc committee of the International Standards Organisation's Technical Committee 172 considered both reference methods for assessing progressive power lenses and which optical characteristics should be readily available to ophthalmic practitioners.

Visual Performance and Related Assessments

There have been few studies of visual performance with different lens designs,²³⁻²⁶ particularly with respect to the relative merits of different base curves. Also the area of perceptual adaptation and spectacle lenses has had little research (apart from work in the area of aniseikonia). Also lacking are practical tasks that require visual-motor coordination (such as orientation and mobility skills) to help choose between different designs.

The field of progressive lens design is the exception to this lack of assessment. There have been many clinical trials. Most of them have been done by manufacturers themselves, and the information is not in the public domain. Vision through progressive lenses is extremely complex and involves many factors that may require more information about perceptual adaptations and the understanding of eye movements during particular tasks.

Future Possibilities

Undoubtedly there will be further developments in aspheric surfaces, both for single-vision and progressive addition lenses. Other possible areas for development include gradient-index lenses, liquid crystal

lenses, and diffraction lenses. Gradient-index lenses could be used to provide single-vision lenses (including simulating affects of aspheric surfaces), bifocals, and progressive addition lenses.^{27,28} Liquid crystal lenses can have their refractive index altered, which will provide an alternative to current progressive addition lenses.²⁹ Bifocal contact lenses and bifocal intraocular lenses are now available that use refraction and zero-order diffraction for the distance correction and refraction and first-order diffraction for the near correction. Possibly this could be extended to bifocal spectacle lenses with the traditional near zone area being available for distance vision also (although with reduced contrast).

There are many technical problems to overcome in these new areas. There are many side effects (including aberrations) associated with these lenses that will keep spectacle lens designers busy for a long time.

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